

NUMERICAL METHODS FINAL EXAM

Date: 21st April, 2025, Time: 10 am - 1 pm, Total Marks 50
Attempt all questions

- (1) Using the complete Horner's algorithm, find the Taylor series of the function $f(x) = 3x^5 - 2x^4 + 15x^3 + 13x^2 - 12x - 5$ at the point $x = 2$. (4 marks)
- (2) Carry out Gaussian elimination with scaled partial pivoting on the following 4×4 matrix and show all intermediate matrices.

$$\begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 3 & -1 \\ 3 & -3 & 0 & 6 \\ 0 & 2 & 4 & -6 \end{bmatrix} \quad (6 \text{ marks})$$

- (3) Prove that two of the four roots of $x^4 + 2x^3 - 7x^2 + 3 = 0$ are positive. Find the approximate values of these roots by using Newton's method. For each root, do two iterations of Newton's method. (2 + 3 + 3 = 8 marks)
- (4) You are given the following data about a function f : $f(0) = 1, f(1) = 9, f(2) = 23, f(4) = 93, f(6) = 259$. Construct the divided difference table for this data and using Newton's interpolation polynomial, find an approximation to $f(4.2)$. (6 marks)
- (5) What is the formula you get by using the composite trapezoid rule on $f(x) = x^2$, with interval $[0, 1]$ and $n + 1$ equally spaced points with $x_0 = 0$ and $x_n = 1$? Simplify your result using the formula for the sum of squares of the first n positive integers. What does this trapezoid estimate converge to as $n \rightarrow \infty$? (4 + 1 + 1 = 6 marks)
- (6) Find a cubic spline S over knots $-1, 0, 1$ such that $S''(-1) = S''(1) = 0, S(-1) = S(1) = 0$ and $S(0) = 1$. (6 marks)
- (7) Solve the differential equation $\frac{dx}{dt} = -tx^2, x(0) = 2$ at $t = 0.2$, correct to two decimal places, using one step of Taylor series method of order 2 and one step of the Runge-Kutta method of order 2. (Help with formulae: The second order Runge-Kutta method is: $x(t + h) = x(t) + (1/2)(K_1 + K_2)$, where $K_1 = hf(t, x)$ and $K_2 = hf(t + h, x + K_1)$). (4 + 4 = 8 marks)
- (8) Show that the following 3×3 matrix cannot be factored into a product of a unit lower triangular matrix and an upper triangular matrix.

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

Interchange the rows of this matrix so that such a factorization can be done. (3 + 3 = 6 marks)